

Theory and Methodology of Science

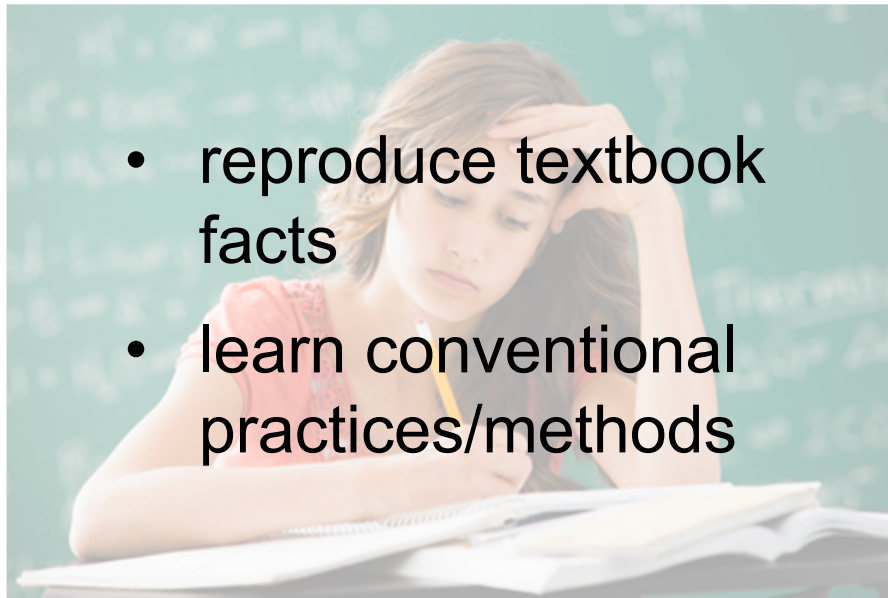
Introduction Lecture

Till Grüne-Yanoff



An Important Transition

Student  Scientist/
Professional/Expert



Method Choice – Examples from MSc Theses



Your task is to identify improvements in the energy performance of a type of buildings. Should you do this by...

FE simulation, or ...
a case study, or...?



You want to investigate wear of a new road cover. Should you do this by...

laboratory experiment, or..
field experiment, or...?

Different Levels of Discussing Method Choice



General philosophy of science

Methodology of science



Methods of science

Different Levels of Discussing Method Choice



General philosophy of science

Methodology of science



Methods of science

FE simulation X-ray spectroscopy Questionnaire
Temperature measurement Pen-and-paper-models
Significance testing Optimization under constraints
Process tracing Regression analysis

Different Levels of Discussing Method Choice



General philosophy of science

How is it possible to acquire knowledge of the world?

Are our best theories true, or are they merely useful?

Different Levels of Discussing Method Choice



General philosophy of science

Methodology of science

- Which of the available methods shall I choose for my goals?
- What justifies this choice of method?

The Focus of TaMoS



Methodology: investigates
assessment and justification of
method choice

Methodology $\neq$ Collection of
methods

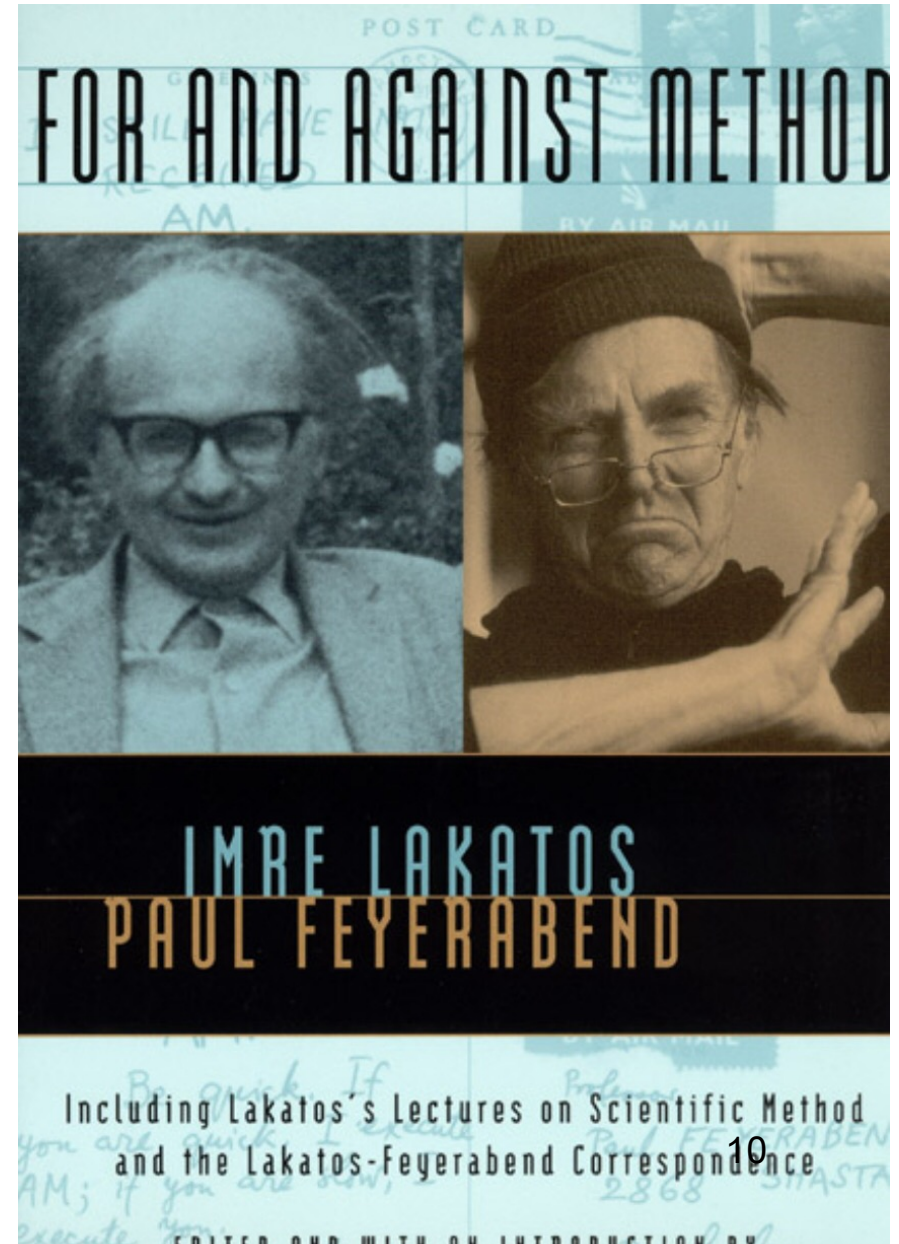


Methods of science

Why *Justify* One's Method Choice?



Why *Justify* One's Method Choice?



Why *Justify* One's Method Choice?

- **Self-reflect** – ensure that you think your choice is the best
- **Plan** – convince your future selves that they should implement your choice
- **Coordinate** – motivate others to join in on your approach
- **Persuade** – give others reasons to trust your results
- **Be accountable** – document your motivations for making this choice
- **Epistemically productive** – there is no knowledge without justification

Practical Matters: The Structure of TaMoS

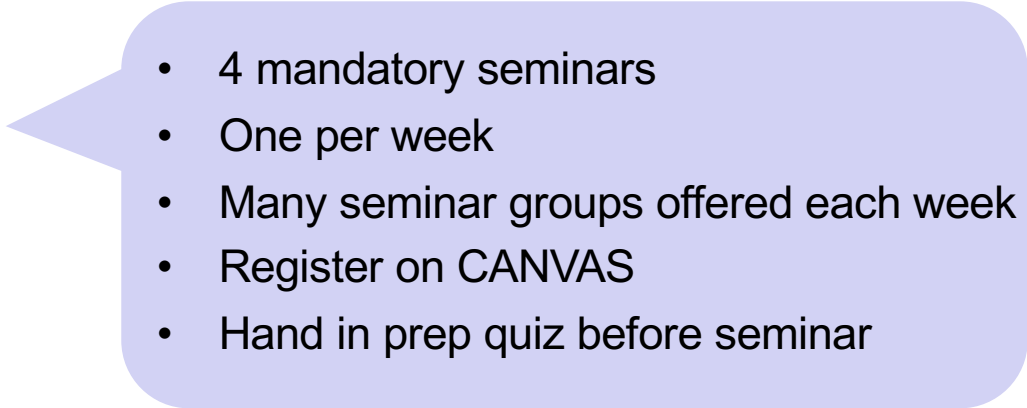
The Structure of TaMoS

- Different course codes – different structure and requirements

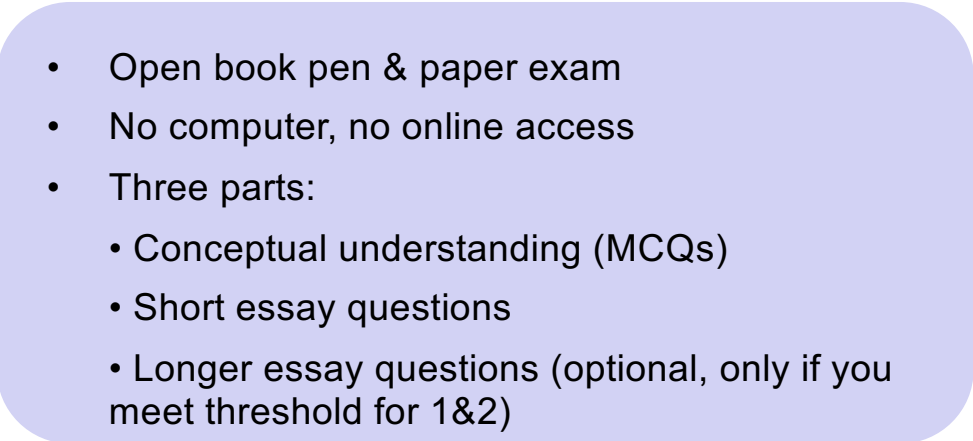
The Structure of TaMoS

- Different course codes – different structure and requirements
- Credit requirements:
 - Prepare for and attend seminars
 - Pass exam
 - (Fulfil project requirements)

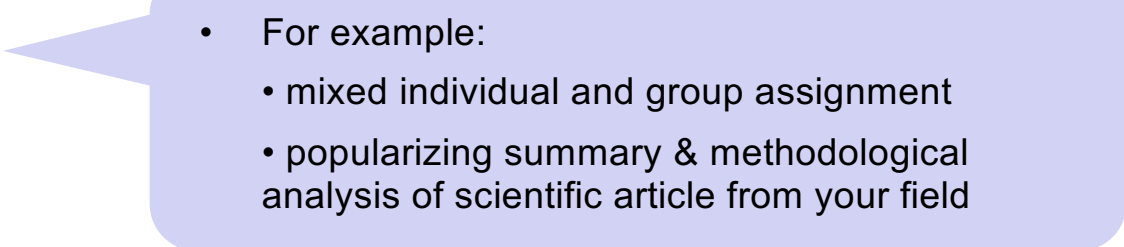
The Structure of TaMoS

- Different course codes – different structure and requirements
- Credit requirements:
 - **Prepare for and attend seminars**
 - 4 mandatory seminars
 - One per week
 - Many seminar groups offered each week
 - Register on CANVAS
 - Hand in prep quiz before seminar
 - Pass exam
 - (Fulfil project requirements)

The Structure of TaMoS

- Different course codes – different structure and requirements
 - Credit requirements:
 - Prepare for and attend seminars
 - **Pass exam**
 - (Fulfil project requirements)
- 
- Open book pen & paper exam
 - No computer, no online access
 - Three parts:
 - Conceptual understanding (MCQs)
 - Short essay questions
 - Longer essay questions (optional, only if you meet threshold for 1&2)

The Structure of TaMoS

- Different course codes – different structure and requirements
 - Credit requirements:
 - Prepare for and attend seminars
 - Pass exam
 - **(Fulfil project requirements)**
- 
- For example:
 - mixed individual and group assignment
 - popularizing summary & methodological analysis of scientific article from your field

The Structure of TaMoS

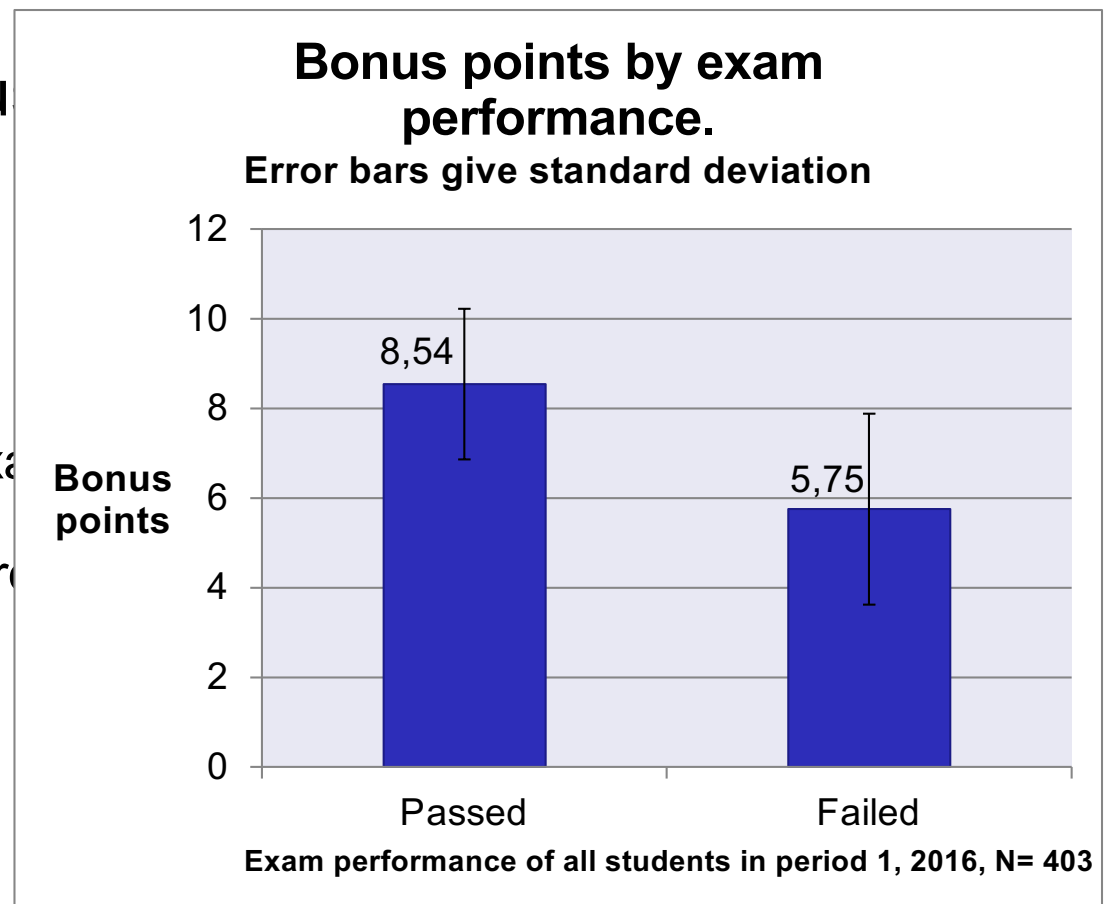
- Different course codes – different structure and requirements
- Credit requirements – seminars, exam, (project)
- Additional offers:
 - Introduction lecture
 - 10 Video lectures (with ca. 5 chapters each) – bonus points for quiz answers
 - 2 Flipped classrooms – bonus points for participation
 - Script covering all video lectures & additional reading material

The Structure of TaMoS

- Different course codes – different structure and requirements
- Credit requirements – seminars, exam, (project)
- Additional offers – campus/video lectures, bonus points
- Bonus points
 - Max 5 points
 - Counts towards part 1 of the exam
 - Good for this term's exam and re-exam

The Structure of TaMoS

- Different course codes – different structure and requirements
- Credit requirements – seminars, exam, (project)
- Additional offers – campus
- Bonus points
 - Max 5 points
 - Counts towards part 1 of the exam
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The Structure of TaMoS

- All information on CANVAS – make sure you have access
 - Schedule
 - Videos
 - Quizz questions
 - Script
 - Old exams
 - Teacher contact

BREAK

How to Justify Method Choice?

Two problems

- The fallibility of justification
- The many meanings of "the scientific method"

How to Justify Method Choice?

Two problems

- The fallibility of justification
- The many meanings of "the scientific method"

Two solutions

- Justification as a social enterprise
- Epistemic tool approach: justify method choice as means to specific epistemic goals

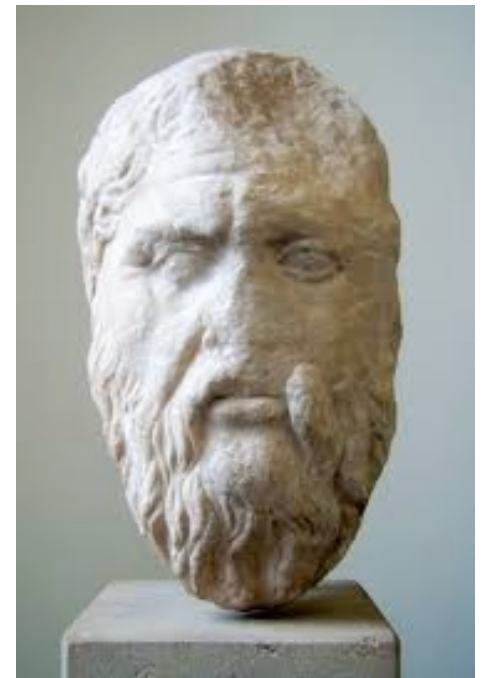
The Nature of Knowledge

Knowledge is true, justified belief

A knows that p only if

- i.* A believes that p .
- ii.* A is justified in believing that p .
- iii.* p is true.

(Not sufficient! Look up [*Gettier problems*](#))



Plato (427-347 BC)

The Nature of Knowledge

Knowledge is true, justified belief

A knows that *p* only if

- i.* *A* believes that *p*.
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- iii.* *p* is true.

- Information that no-one believes to be true cannot be knowledge
- All knowledge is held by agents who act based on their beliefs
- Beliefs and hence knowledge can be uncertain

The Nature of Knowledge

Knowledge is true, justified belief

A knows that *p* only if

- i.* *A* believes that *p*.
- ii.* *A* is justified in believing that *p*.
- iii.* *p* is true.

- You might accidentally believe the truth – that is *not* knowledge
- Knowing requires providing valid reasons for your beliefs
- That includes providing valid reasons for having chosen the methods that provide evidence

The Nature of Knowledge

Knowledge is true, justified belief

A knows that *p* only if

- i.* *A* believes that *p*.
- ii.* *A* is justified in believing that *p*.
- iii.* *p* is true.

If the claim *p* is false, then it cannot be knowledge...

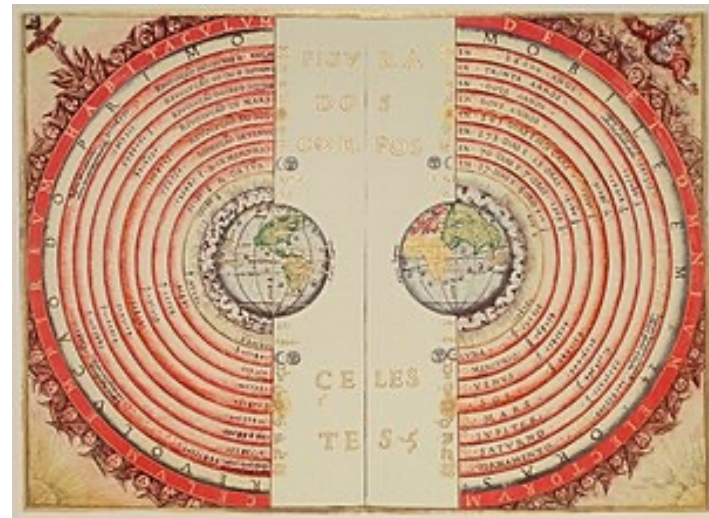
...even if you believe it to be true

...even if your belief in its truth is justified

The Fallibility of Justification

If a belief is justified, it does *not* necessarily follow that this belief is true

Tycho Brahe
1546-1601

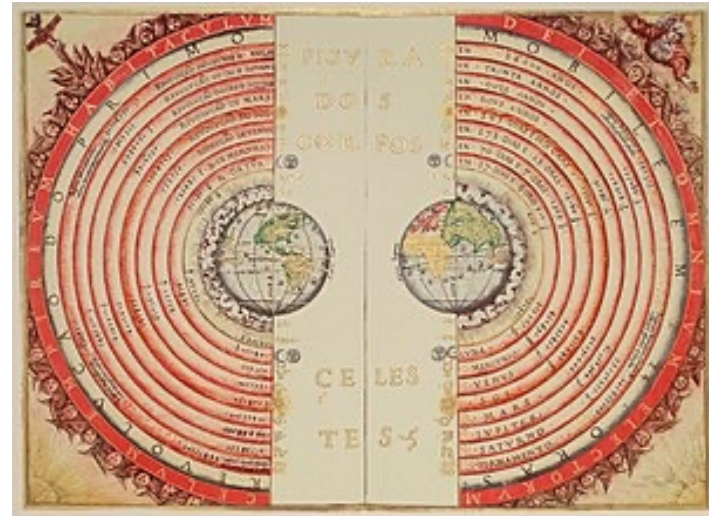


Geocentric model of
the universe by B.
Velho, 1568

The Fallibility of Justification

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Example:

Astronomy before the telescope

Best available sensory experience did not suffice to justify rejection of Geocentric model



Geocentric model of the universe by B. Velho, 1568

Galileo's telescope, c. 1610

The Fallibility of Justification

Uncertain knowledge – claims that are true and justified, but where justification is not sufficient to believe claim with certainty

Fallible justifications – better justified claims tend to be true, but their truth is not guaranteed

Fallible methods – methods that tend to give valid justifications, but there is a chance that they go wrong

“The Scientific Method”?

“The scientific method is an empirical method for acquiring knowledge that has been referred to while doing science since at least the 17th century.” (Wikipedia)

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“The scientific method is an empirical method for acquiring knowledge that has been referred to while doing science since at least the 17th century.” (Wikipedia)

Perhaps fallibility of justification can be minimized by adhering to a universal, fixed set of prescribed steps (a universal algorithm)?

“The Scientific Method”?

Issue 1: Diversity of science

- Different goals
- Different methods
- Different standards

“The Scientific Method”?

Issue 2: Lack of an accurate demarcation criteria

“The Scientific Method”?

Issue 2: Lack of an accurate demarcation criteria

- e.g. Popper’s falsifiability criterion

“statements... in order to be ranked as scientific, must be capable of conflicting with possible, or conceivable observations.”

Epistemic Tool Justification

Choose the method that tends to best satisfy your epistemic goals

Epistemic Tool Justification

Choose the method that tends to best satisfy your specific epistemic goal

- Identify actual epistemic goal
- Conceptual analysis: what properties important for that goal?
- Instrumental rationality: choose methods that best satisfy these properties, given available knowledge

Epistemic Tool Justification

Choose the method that tends to best satisfy your epistemic goals

- Reasons: considerations that justify choosing a method (over others) for a given goal

Method A	
<i>pro</i>	<i>con</i>
...	...

Method B	
<i>pro</i>	<i>con</i>
...	...

- Choose the method that has the stronger supporting or the weaker contradicting reasons

Epistemic Tool Justification - Illustration



Your task is to investigate wear of a new road cover.

Should you do this by...

- laboratory experiment, or field experiment, or observational study?

Epistemic Tool Justification - Illustration



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Goal: *explain* why the new cover wears differently than the previous one in Sweden

Analysis: explanation requires identifying the difference-making causes

Epistemic Tool Justification - Illustration



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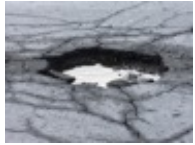
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Comparison:

Lab experiment	Field experiment	Observational study
Good tool to identify causes	Good tool to identify causes	Less good tool to identify causes

Epistemic Tool Justification - Illustration



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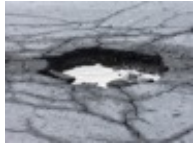
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Comparison:

Lab experiment	Field experiment	Observational study
Good tool to identify causes	Good tool to identify causes	Less good tool to identify causes
Good background variable control	Less good background variable control	Less good background variable control
Difficult to generate actual background conditions	Set in actual background conditions	Set in actual background conditions

Learning Results

- Distinguish methodology from methods
- Discuss the importance of justifying method choice
- Explain the classical definition of knowledge
- Describe the concept of belief and the fallibility of justification
- Explain why there is no universal scientific method prescriptive algorithm
- Explain the epistemically tool approach to method choice justification